

Tien-Sheng Chao

professor

基本信息

姓名: Tien-Sheng Chao

所属院校: 台湾阳明交通大学

地区: China-Taiwan

年龄范围: 50岁以上

学科领域

微电子科学与工程

材料物理

集成电路设计与集成系统

电子信息材料

研究领域

微电子器件与制造

纳米电子与先进工艺

薄膜材料与表面物理

半导体与光电子材料

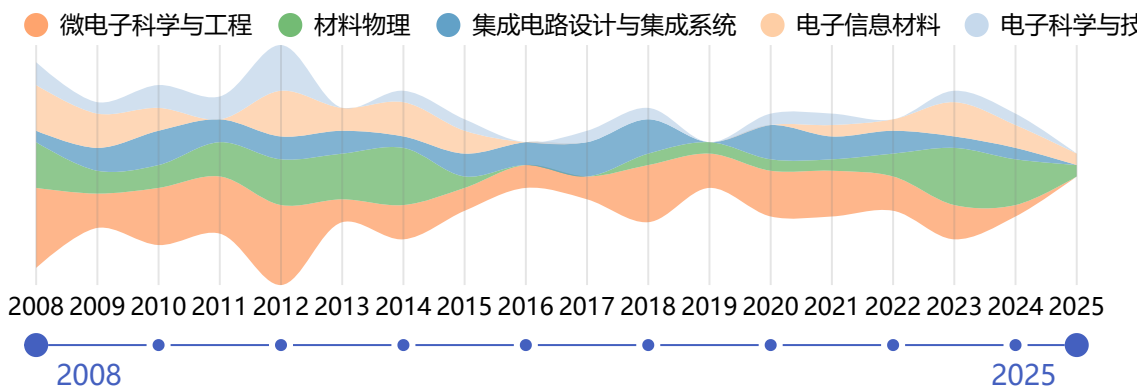
半导体材料与器件

薄膜与表面技术

个人简介

Tien-Sheng Chao is a professor in the Degree Program of Science at NCTU. He holds a PhD in Electronics Engineering from National Yang Ming Chiao Tung University. His research interests include Physics of Semiconductor Devices, Thin Films Deposition Techniques, etc. He has rich experience, serving as an associate researcher at National Nano Device Labs from 1992 - 2001, and then an associate professor and professor at NCTU. He was also the chairman of the Department of Electrophysics at NCTU from 2009 - 2011. With 3670 citations and an h - index of 28, he has 24 projects, 371 research outputs, and 7 prizes. His ORCID is <https://orcid.org/0000-0001-9618-207X>, phone: 03 - 5131367, and email: tschao.

近年发表的论文



In Situ O₂ Plasma-Treated HfO₂ -ZrO₂ Superlattice HZO FeRAMs Exhibiting Enhanced Remnant Polarization and Improved Endurance Performance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hsieh, Dong-Ru; Hong, Zi-Yang; Yeh, Wei-Ju; Ni, Jia-Chian; Luo, Huai-En; Liang, Yan-Kui; Lin, Chun-Hsiung; Chao, Tien-Sheng

被引用次数: 0 2025

Investigation on Germanium Transistors with HfO_x-Based Gate Stacks for CMOS Operations

ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY

Hong, Tzu-Chieh; Su, Chun-Jung; Lee, Yao-Jen; Li, Yiming; Samukawa, Seiji; Chao, Tien-Sheng

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Improved Stability of BEOL-Compatible Highly Scaled Ultrathin InZnO Channel Ferroelectric Thin-Film Transistor With TiO₂ Interfacial Layer

IEEE TRANSACTIONS ON ELECTRON DEVICES

Liang, Yan-Kui; Li, Wei-Li; Lin, Yu-Lon; Hsieh, Dong-Ru; Yang, Tsung-Ying; Chou, Tsung-Te; Kei, Chi-Chung; Huang, Huai-Ying; Lin, Yu-Ming; Tseng, Yuan-Chieh; Chao, Tien-Sheng; Chang, Edward Yi; Toprasertpong, Kasidit; Takagi, Shinichi; Lin, Chun-Hsiung

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Antiferroelectric Hf_{0.25}Zr_{0.75}O₂ With Ferroelectric Properties and Low Voltage Operation of AFeRAM and AFeFET by Using Low-Temperature Atomic Layer Deposition

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Kun-Tao; Lo, Chieh; Chang, Shu-Chieh; Tsai, Hsing-Tzu; Chao, Tien-Sheng

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2024 IEEE SILICON NANOELECTRONICS WORKSHOP, SNW 2024

Liang, Yan-Kui; Peng, Li-Chi; Lin, Yu-Lon; Zheng, June-Yang; Hsieh, Dong-Ru; Chou, Tsung-Te; Huang, Huai-Ying; Lin, Yu-Ming; Tseng, Yuan-Chieh; Chao, Tien-Sheng; Chang, Edward-Yi; Toprasertpong, Kasidit; Takagi, Shinichi; Lin, Chun-Hsiung

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Investigation of HfO₂ and ZrO₂ Nanolamination Thickness on Superlattice HZO FeRAMs Exhibiting Enhanced Remnant Polarization and Improved Endurance

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2024

Single-Gate In-Transistor Readout of Current Superposition and Collapse Utilizing Quantum Tunneling and Ferroelectric Switching

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Chen, Ching-Hung; Lai, Yu-Ting; Chen, Ciao-Fen; Wu, Pei-Tzu; Su, Kuan-Jung; Hsu, Sheng-Yang; Dai, Guo-Jin; Huang, Zan-Yi; Hsu, Chien-Lung; Lee, Shen-Yang; Shen, Chuan-Hui; Chen, Hsin-Yu; Lee, Chia-Chin; Hsieh, Dong-Ru; Lin, Yen-Fu; Chao, Tien-Sheng; Lo, Shun-Tsung

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A high thermal stability ohmic contact for GaN-based devices

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Wu, Chia-Yi; Chao, Tien-Sheng; Chou, Yi-Chia

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Highly Stable Short Channel Ultrathin Atomic Layer Deposited Indium Zinc Oxide Thin Film Transistors With Excellent Electrical Characteristics

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Liang, Yan-Kui; Li, Wei-Li; Zheng, Jun-Yang; Lin, Yu-Lon; Lu, Yu-Cheng; Chiu, Ching-Hua; Hsieh, Dong-Ru; Chou, Tsung-Te; Kei, Chi-Chung; Huang, Huai-Ying; Lin, Yu-Ming; Tseng, Yuan-Chieh; Chao, Tien-Sheng; Chang, Edward Yi; Lin, Chun-Hsiung

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Reliability of Multiple-Layer Stacked Gate-All-Around Poly-Si Nanosheet Channel Ferroelectric HfxZr1-xO2 FETs With NH3 Plasma Treatment

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SEMICONDUCTOR SCIENCE AND TECHNOLOGY

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A Harmonic Radar Tag With High Detection Range Utilizing Ge FinFETs CMOS Technology

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High Endurance and Low Fatigue Effect of Bilayer Stacked Antiferroelectric/Ferroelectric HfxZr1-xO2

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Role of Nitrogen in Ferroelectricity of HfxZr1-xO2-Based Capacitors With Metal-Ferroelectric-Insulator-Metal Structure

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First Demonstration of Heterogeneous IGZO/Si CFET Monolithic 3-D Integration With Dual Work Function Gate for Ultralow-Power SRAM and RF Applications

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Chang, Shu-Wei; Lu, Tsung-Han; Yang, Cong-Yi; Yeh, Cheng-Jui; Huang, Min-Kun; Meng, Ching-Fan; Chen, Po-Jen; Chang, Ting-Hsuan; Chang, Yan-Shiuan; Jhu, Jhe-Wei; Hong, Tzu-Chieh; Ke, Chu-Chu; Yu, Xin-Ren; Lu, Wen-Hsiang; Baig, Mohammed Aftab; Cho, Ta-Chun; Sung, Po-Jung; Su, Chun-Jung; Hsueh, Fu-Kuo; Chen, Bo-Yuan; Hu, Hsin-Hui; Wu, Chien-Ting; Lin, Kun-Lin; Ma, William Cheng-Yu; Lu, Darsen D.; Kao, Kuo-Hsing; Lee, Yao-Jen; Lin, Cheng-Li; Huang, Kun-Ping; Chen, Kun-Ming; Li, Yiming; Samukawa, Seiji; Chao, Tien-Sheng; Huang, Guo-Wei; Wu, Wen-Fa; Lee, Wen-Hsi; Li, Jiun-Yun; Shieh, Jia-Min; Tarn, Jenn-Hwan; Wang, Yeong-Her; Yeh, Wen-Kuan

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Lee, Shen-Yang; Lee, Chia-Chin; Kuo, Yi-Shan; Li, Shou-Wei; Chao, Tien-Sheng

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Fabrication of Vertically Stacked Nanosheet Junctionless Field-Effect Transistors and Applications for the CMOS and CFET Inverters

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Effect of Seed Layer on Gate-All-Around Poly-Si Nanowire Negative-Capacitance FETs With MFMIS and MFIS Structures: Planar Capacitors to 3-D FETs

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Effects of Forming Gas Annealing and Channel Dimensions on the Electrical Characteristics of FeFETs and CMOS Inverter

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Characteristics of Poly-Si Junctionless FinFETs With HfZrO Using Forming Gas Annealing

IEEE TRANSACTIONS ON NANOTECHNOLOGY

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Nitride Induced Stress Affecting Crystallinity of Sidewall Damascene Gate-All-Around Nanowire Poly-Si FETs

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Characteristics of In_{0.7}Ga_{0.3}As MOS Capacitors Obtained using Hydrochloric Acid Treatment, Ammonium Sulfide Passivation, Methanol Treatment, and Forming Gas Annealing

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Chung, Sheng-Ti; Huang, Yi-Chin; Fu, Yen-Chun; Lee, Yao-Jen; Chao, Tien-Sheng

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Experimental Demonstration of Stacked Gate- All-Around Poly-Si Nanowires Negative Capacitance FETs With Internal Gate Featuring Seed Layer and Free of Post-Metal Annealing Process

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Lee, Shen-Yang; Chen, Han-Wei; Shen, Chiuan-Huei; Kuo, Po-Yi; Chung, Chun-Chih; Huang, Yu-En; Chen, Hsin-Yu; Chao, Tien-Sheng

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2019

Self-Limited Low-Temperature Trimming and Fully Silicided S/D for Vertically Stacked Cantilever Gate-All-Around Poly-Si Junctionless Nanosheet Transistors

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Investigation of Nitrous Oxide Nitridation Temperatures on P-Type Pi-Gate Poly-Si Junctionless Accumulation Mode TFTs

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Hsieh, Dong-Ru; Lin, Kun-Cheng; Chao, Tien-Sheng

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2019

Variable-Channel Junctionless Poly-Si FETs: Demonstration and Investigation With Different Body Doping Concentrations

IEEE ELECTRON DEVICE LETTERS

Lin, Jer-Yi; Tsai, Chan-Yi; Shen, Chiuan-Huei; Chung, Chun-Chih; Kumar, Malkundi Puttaveerappa Vijay; Chao, Tien-Sheng

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Stacked Sidewall-Damascene Double-Layer Poly-Si Trigate FETs With RTA-Improved Crystallinity

IEEE ELECTRON DEVICE LETTERS

Shen, Chiuan-Huei; Kuo, Po-Yi; Chung, Chun-Chih; Lee, Sen-Yang; Chao, Tien-Sheng

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2018

Junctionless Nanosheet (3 nm) Poly-Si TFT: Electrical Characteristics and Superior Positive Gate Bias Stress Reliability

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Lin, Jer-Yi; Kumar, Malkundi Puttaveerappa Vijay; Chao, Tien-Sheng

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Junctionless FETs With a Fin Body for Multi-VTH and Dynamic Threshold Operation

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2018

Improving the Electrical Performance of a Quantum Well FET With a Shell Doping Profile by Heterojunction Optimization

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Kumar, Malkundi Puttaveerappa Vijay; Hu, Chia-Ying; Walke, Amey Mahadev; Kao, Kuo-Hsing; Chao, Tien-Sheng

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Comprehensive Analysis on Electrical Characteristics of Pi-Gate Poly-Si Junctionless FETs

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High-Performance Uniaxial Tensile Strained n-Channel JL SOI FETs and Triangular JL Bulk FinFETs for Nanoscaled Applications

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Sung, Po-Jung; Cho, Ta-Chun; Hou, Fu-Ju; Hsueh, Fu-Kuo; Chung, Sheng-Ti; Lee, Yao-Jen; Current, Michael I.; Chao, Tien-Sheng

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High-performance sidewall damascened tri-gate poly-si TFTs with the strain proximity free technique and stress memorization technique

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Hsieh, Dong-Ru; Kuo, Po-Yi; Lin, Jer-Yi; Chen, Yi-Hsuan; Chang, Tien-Shun; Chao, Tien-Sheng

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2017

Label-free and real-time detection of ferritin using a horn-like polycrystalline-silicon nanowire field-effect transistor biosensor

SENSORS AND ACTUATORS B-CHEMICAL

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Junctionless Poly-Si Nanowire Transistors With Low-Temperature Trimming Process for Monolithic 3-D IC Application

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Switching characteristics in Cu:SiO₂ by chemical soak methods for resistive random access memory (ReRAM)

SOLID-STATE ELECTRONICS

Chin, Fun-Tat; Lin, Yu-Hsien; Yang, Wen-Luh; Liao, Chin-Hsuan; Lin, Li-Min; Hsiao, Yu-Ping; Chao, Tien-Sheng

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2015

Impact of Crystallization Method on Poly-Si Tunnel FETs

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Chen, Yi-Hsuan; Ma, William Cheng-Yu; Lin, Jer-Yi; Lin, Chun-Yen; Hsu, Po-Yang; Huang, Chi-Yuan; Chao, Tien-Sheng

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High-performance poly-Si TFT with ultra-thin channel film and gate oxide for low-power application

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Chen, Yi-Hsuan; Ma, William Cheng-Yu; Chao, Tien-Sheng

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Advanced Cu chemical displacement technique for SiO₂-based electrochemical metallization ReRAM application

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Chin, Fun-Tat; Lin, Yu-Hsien; You, Hsin-Chiang; Yang, Wen-Luh; Lin, Li-Min; Hsiao, Yu-Ping; Ko, Chum-Min; Chao, Tien-Sheng

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2014

Characterization of Ultra-Thin Ni Silicide Film by Two-Step Low Temperature Microwave Anneal

ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY

Wu, Chien-Ting; Lee, Yao-Jen; Hsueh, Fu-Kuo; Sung, Po-Jung; Cho, Ta-Chun; Current, Michael Ira; Chao, Tien-Sheng

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Effect of Sensing Film Thickness on Sensing Characteristics of Dual-Gate Poly-Si Ion-Sensitive Field-Effect-Transistors

IEEE ELECTRON DEVICE LETTERS

Yen, Li-Chen; Tang, Ming-Tsyr; Tan, Chia-Ying; Pan, Tung-Ming; Chao, Tien-Sheng

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High-Performance GAA Sidewall-Damascened Sub-10-nm In Situ n⁺-Doped Poly-Si NWs Channels Junctionless FETs

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Ion-Bombarded and Plasma-Passivated Charge Storage Layer for SONOS-Type Nonvolatile Memory

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Low-Temperature Microwave Annealing Processes for Future IC Fabrication-A Review

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Lee, Yao-Jen; Cho, Ta-Chun; Chuang, Shang-Shiun; Hsueh, Fu-Kuo; Lu, Yu-Lun; Sung, Po-Jung; Chen, Hsiu-Chih; Current, Michael I.; Tseng, Tseung-Yuen; Chao, Tien-Sheng; Hu, Chenming; Yang, Fu-Liang

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Improvement in pH Sensitivity of Low-Temperature Polycrystalline-Silicon Thin-Film Transistor Sensors Using H₂ Sintering

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Yen, Li-Chen; Tang, Ming-Tsyr; Chang, Fang-Yu; Pan, Tung-Ming; Chao, Tien-Sheng; Lee, Chiang-Hsuan

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Microwave Annealing of Phosphorus and Cluster Carbon Implanted (100) and (110) Si

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Cho, Ta-Chun; Lu, Yu-Lun; Yao, Jie-Yi; Lee, Yao-Jen; Sekar, Karuppanan; Tokoro, Nobuhiro; Onoda, Hiroshi; Krull, Wade; Current, Michael I.; Chao, Tien-Sheng

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2013

Novel Ion Bombardment Technique for Doping Limited Cu Source in SiO_x-Based Nonvolatile Switching Layer

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Liu, Sheng-Hsien; Yang, Wen-Luh; Lin, Yu-Hsien; Wu, Chi-Chang; Chao, Tien-Sheng

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Improved Rear-Side Passivation by Atomic Layer Deposition Al₂O₃/SiN_x Stack Layers for High VOC Industrial p-Type Silicon Solar Cells

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Lin, Je-Wei; Chen, Yi-Yang; Gan, Jon-Yiew; Hseih, Wei-Ping; Du, Chen-Hsu; Chao, Tien-Sheng

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Low-Temperature Polycrystalline-Silicon Tunneling Thin-Film Transistors With MILC

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Chen, Yi-Hsuan; Yen, Li-Chen; Chang, Tien-Shun; Chiang, Tsung-Yu; Kuo, Po-Yi; Chao, Tien-Sheng

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Channel Thickness Effect on High-Frequency Performance of Poly-Si Thin-Film Transistors

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Enhancement of Open-Circuit Voltage Using CF₄ Plasma Treatment on Nitric Acid Oxides

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Impacts of Multiple Strain-Gate Engineering on a Zero-Temperature-Coefficient Point

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High-Performance Polyimide-Based ReRAM for Nonvolatile Memory Application

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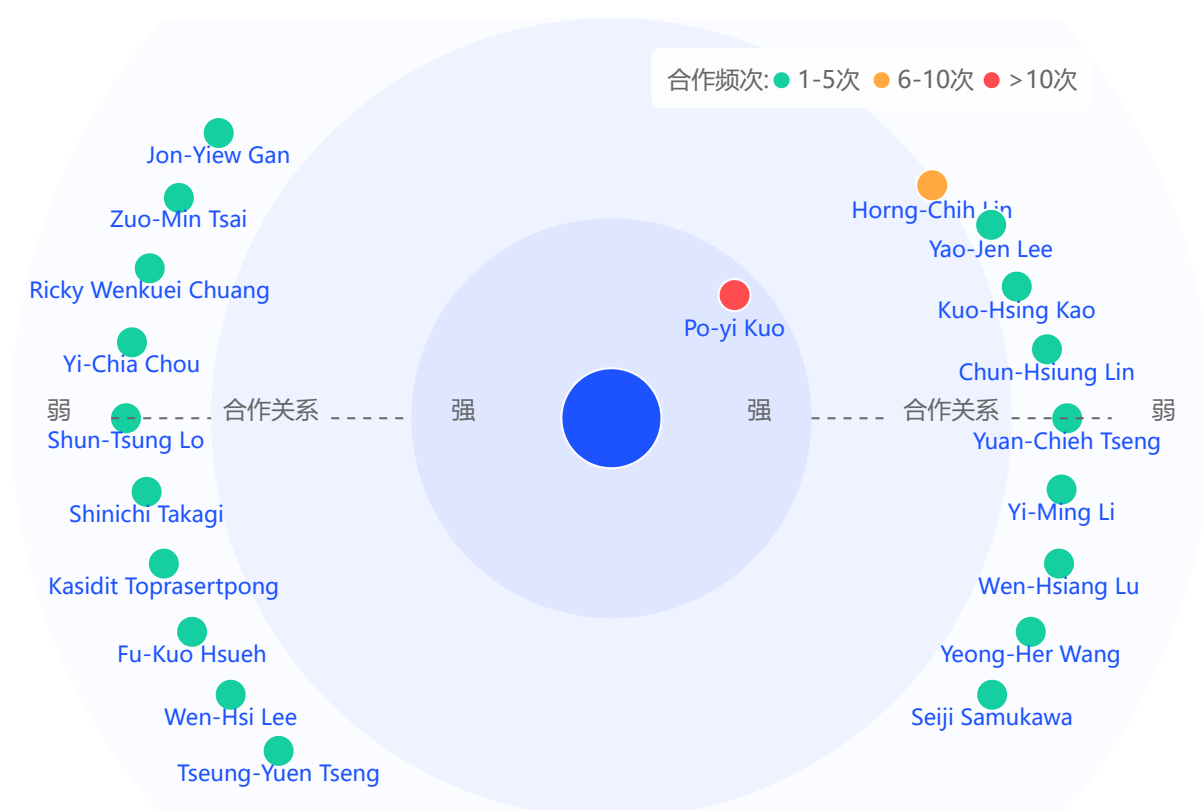
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台湾阳明交通大学	
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Kuo-Hsing Kao	合作次数: 5
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Chun-Hsiung Lin	合作次数: 4
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